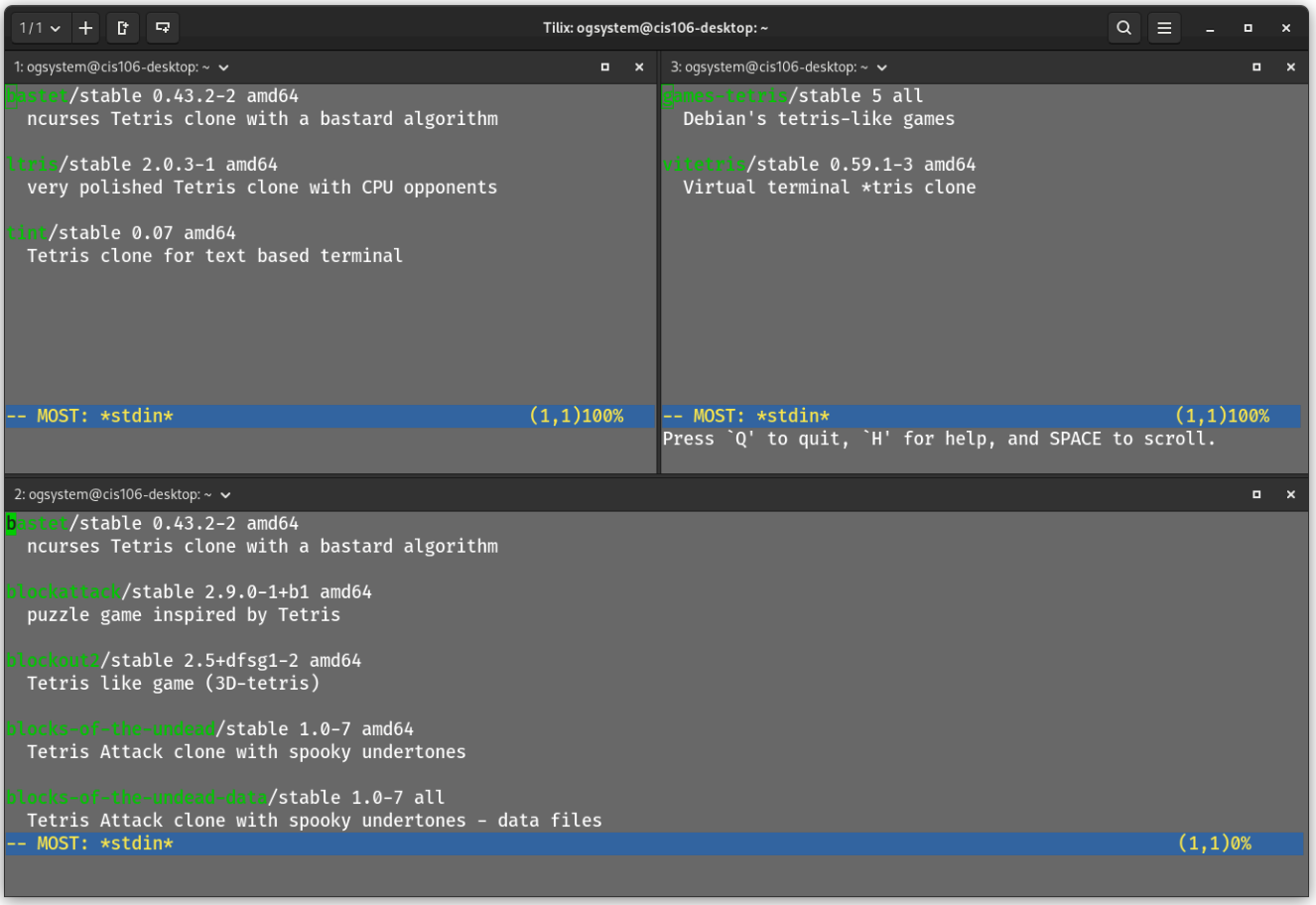
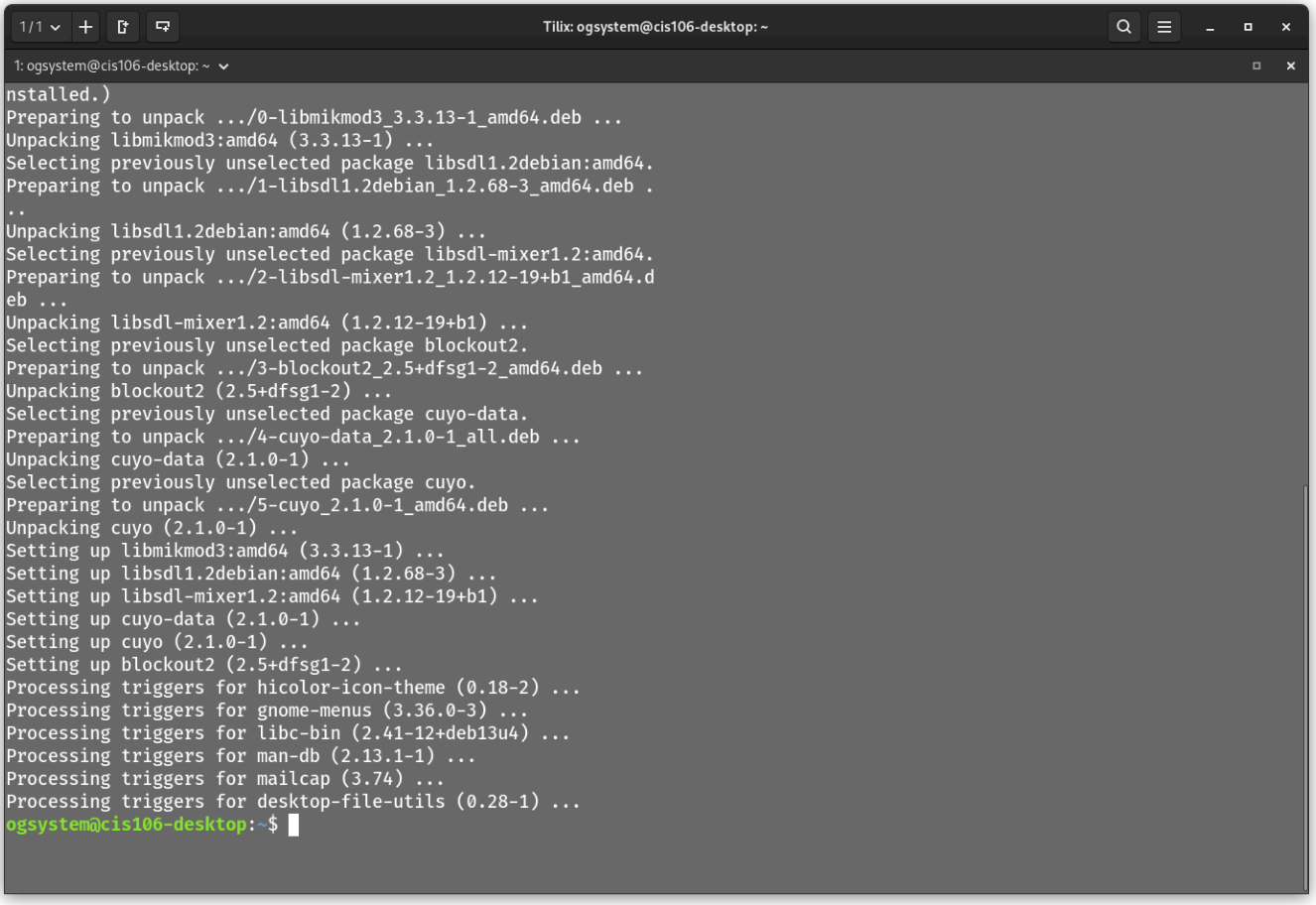
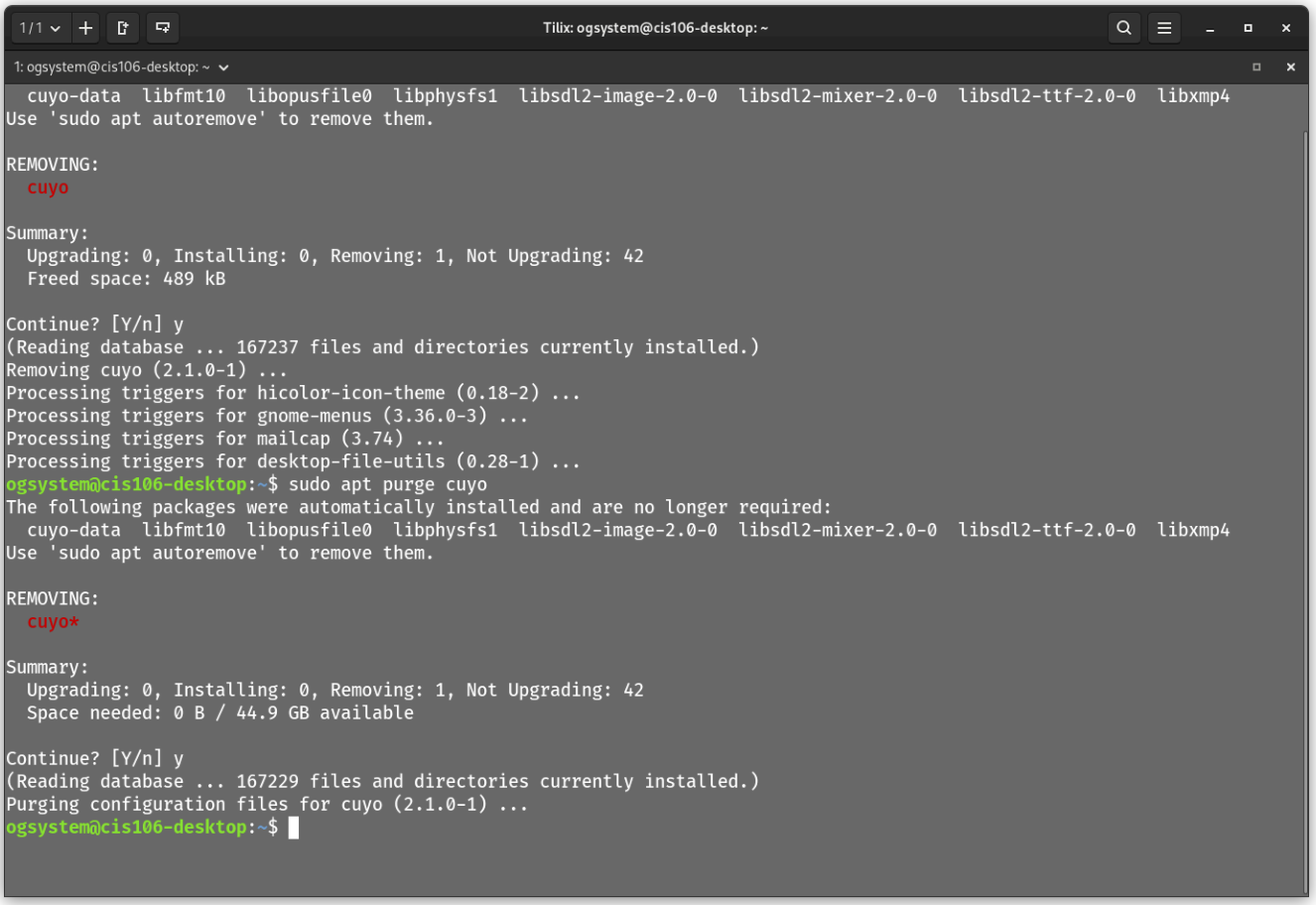
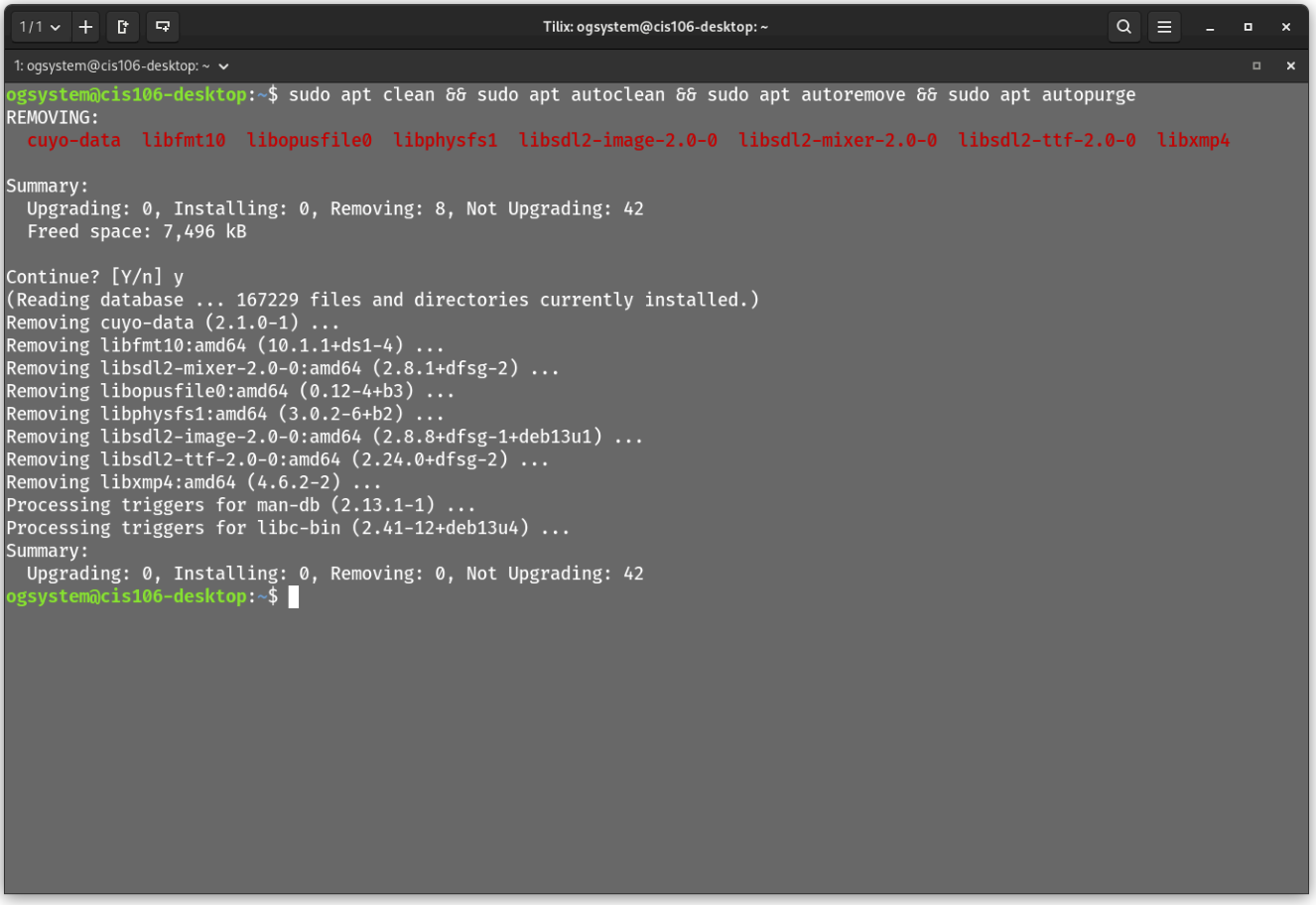
Lab 4 Submission

Question 1

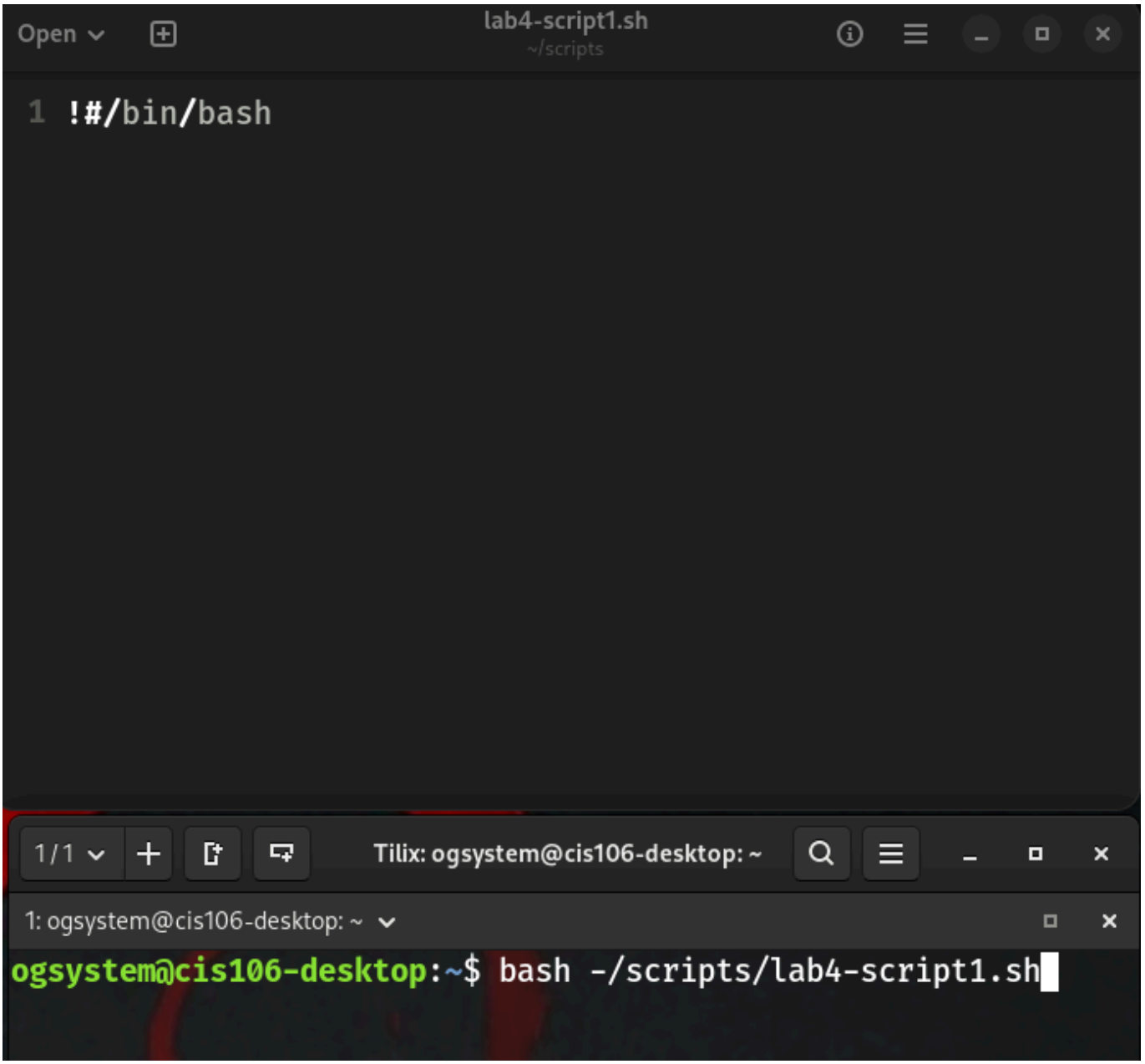

Tilix: ogsystem@cis106-desktop: ~
1: ogsystem@cis106-desktop: ~
3: ogsystem@cis106-desktop: ~
bastet/stable 0.43.2-2 amd64
ncurses Tetris clone with a bastard algorithm
ltris/stable 2.0.3-1 amd64
very polished Tetris clone with CPU opponents
tint/stable 0.07 amd64
Tetris clone for text based terminal
-- MOST: *stdin* (1,1)100%
games-tetris/stable 5 all
Debian's tetris-like games
vitetris/stable 0.59.1-3 amd64
Virtual terminal *tris clone
-- MOST: *stdin* (1,1)100%
Press `Q' to quit, `H' for help, and SPACE to scroll.
2: ogsystem@cis106-desktop: ~
bastet/stable 0.43.2-2 amd64
ncurses Tetris clone with a bastard algorithm
blockattack/stable 2.9.0-1+b1 amd64
puzzle game inspired by Tetris
blockout2/stable 2.5+dfsg1-2 amd64
Tetris like game (3D-tetris)
blocks-of-the-undead/stable 1.0-7 amd64
Tetris Attack clone with spooky undertones
blocks-of-the-undead-data/stable 1.0-7 all
Tetris Attack clone with spooky undertones - data files
-- MOST: *stdin* (1,1)0%

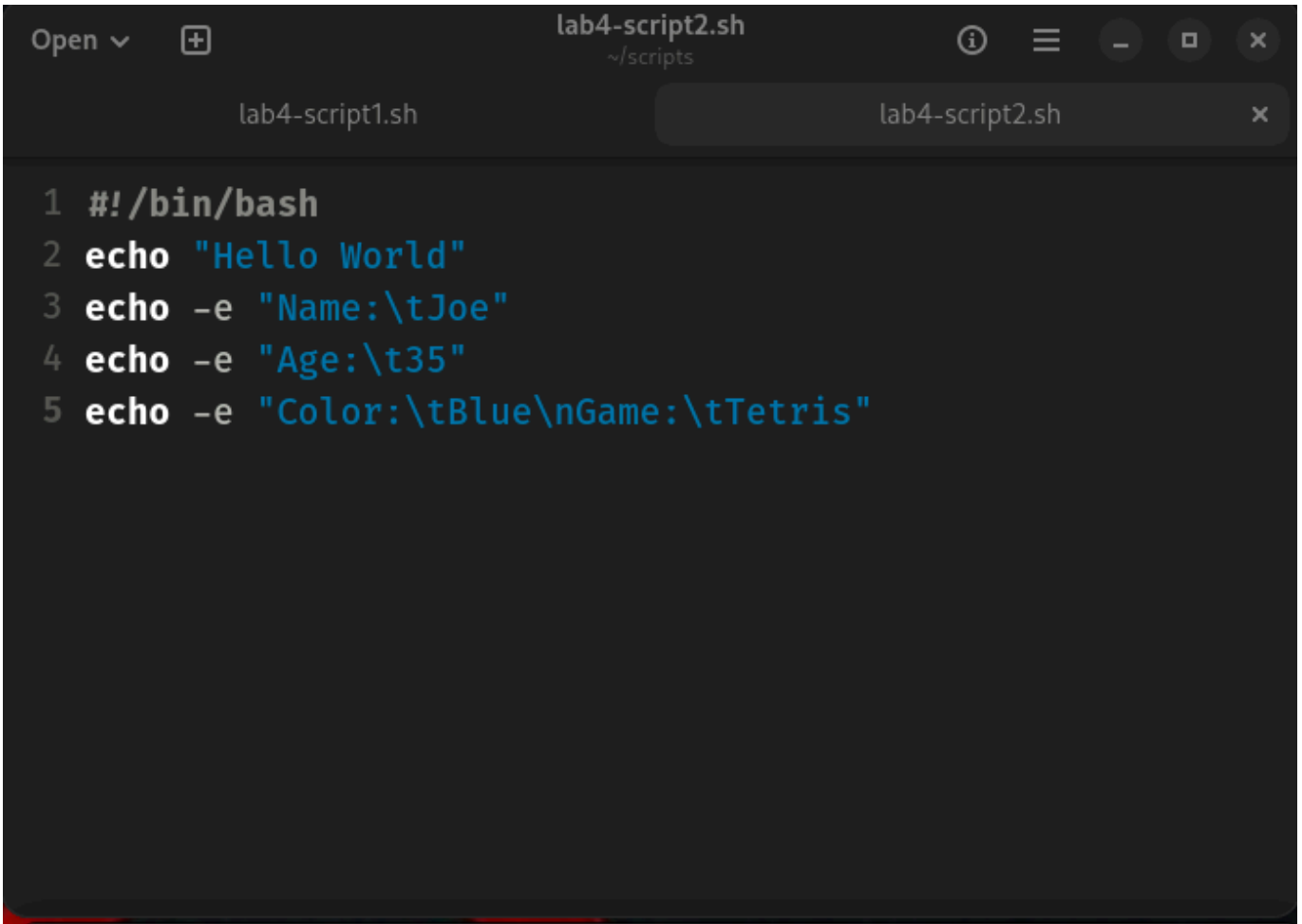

Tilix: ogsystem@cis106-desktop: ~
1: ogsystem@cis106-desktop: ~
nstalled.)
Preparing to unpack .../0-libmikmod3_3.3.13-1_amd64.deb ...
Unpacking libmikmod3:amd64 (3.3.13-1) ...
Selecting previously unselected package libsdl1.2debian:amd64.
Preparing to unpack .../1-libsdl1.2debian_1.2.68-3_amd64.deb .
..
Unpacking libsdl1.2debian:amd64 (1.2.68-3) ...
Selecting previously unselected package libsdl-mixer1.2:amd64.
Preparing to unpack .../2-libsdl-mixer1.2_1.2.12-19+b1_amd64.d
eb ...
Unpacking libsdl-mixer1.2:amd64 (1.2.12-19+b1) ...
Selecting previously unselected package blockout2.
Preparing to unpack .../3-blockout2_2.5+dfsg1-2_amd64.deb ...
Unpacking blockout2 (2.5+dfsg1-2) ...
Selecting previously unselected package cuyo-data.
Preparing to unpack .../4-cuyo-data_2.1.0-1_all.deb ...
Unpacking cuyo-data (2.1.0-1) ...
Selecting previously unselected package cuyo.
Preparing to unpack .../5-cuyo_2.1.0-1_amd64.deb ...
Unpacking cuyo (2.1.0-1) ...
Setting up libmikmod3:amd64 (3.3.13-1) ...
Setting up libsdl1.2debian:amd64 (1.2.68-3) ...
Setting up libsdl-mixer1.2:amd64 (1.2.12-19+b1) ...
Setting up cuyo-data (2.1.0-1) ...
Setting up cuyo (2.1.0-1) ...
Setting up blockout2 (2.5+dfsg1-2) ...
Processing triggers for hicolor-icon-theme (0.18-2) ...
Processing triggers for gnome-menus (3.36.0-3) ...
Processing triggers for libc-bin (2.41-12+deb13u4) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for mailcap (3.74) ...
Processing triggers for desktop-file-utils (0.28-1) ...
ogsystem@cis106-desktop:~$

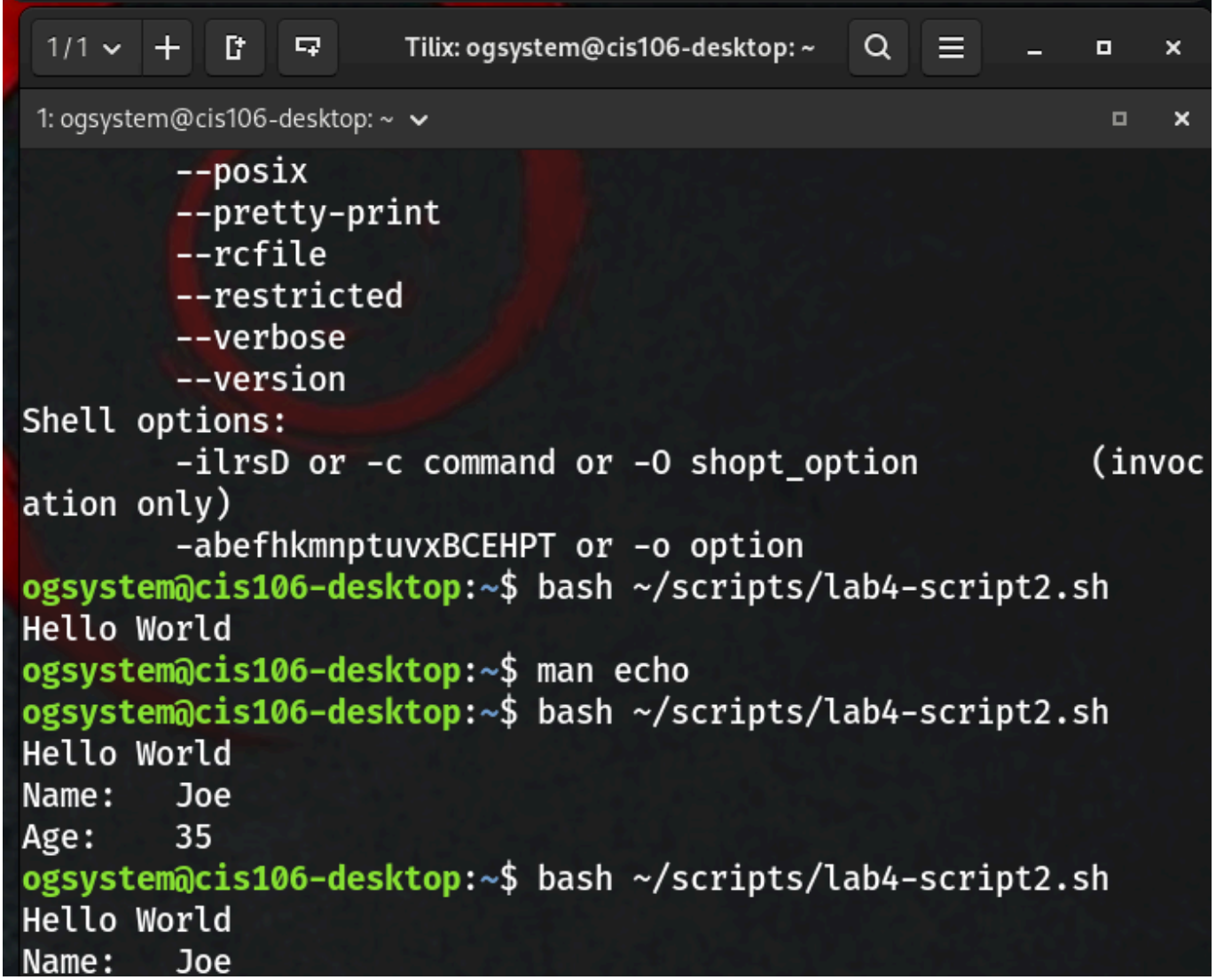

1 / 1
Tilix: ogsystem@cis106-desktop: ~
1: ogsystem@cis106-desktop: ~
cuyo-data libfmt10 libopusfile0 libphysfs1 libsdl2-image-2.0-0 libsdl2-mixer-2.0-0 libsdl2-ttf-2.0-0 libxmp4
Use 'sudo apt autoremove' to remove them.
REMOVING:
cuyo
Summary:
Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 42
Freed space: 489 kB
Continue? [Y/n] y
(Reading database ... 167237 files and directories currently installed.)
Removing cuyo (2.1.0-1) ...
Processing triggers for hicolor-icon-theme (0.18-2) ...
Processing triggers for gnome-menus (3.36.0-3) ...
Processing triggers for mailcap (3.74) ...
Processing triggers for desktop-file-utils (0.28-1) ...
ogsystem@cis106-desktop:~$ sudo apt purge cuyo
The following packages were automatically installed and are no longer required:
cuyo-data libfmt10 libopusfile0 libphysfs1 libsdl2-image-2.0-0 libsdl2-mixer-2.0-0 libsdl2-ttf-2.0-0 libxmp4
Use 'sudo apt autoremove' to remove them.
REMOVING:
cuyo*
Summary:
Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 42
Space needed: 0 B / 44.9 GB available
Continue? [Y/n] y
(Reading database ... 167229 files and directories currently installed.)
Purging configuration files for cuyo (2.1.0-1) ...
ogsystem@cis106-desktop:~$


1 / 1
Tilix: ogsystem@cis106-desktop: ~
1: ogsystem@cis106-desktop: ~
ogsystem@cis106-desktop:~$ sudo apt clean && sudo apt autoclean && sudo apt autoremove && sudo apt autopurge
REMOVING:
cuyo-data libfmt10 libopusfile0 libphysfs1 libsdl2-image-2.0-0 libsdl2-mixer-2.0-0 libsdl2-ttf-2.0-0 libxmp4
Summary:
Upgrading: 0, Installing: 0, Removing: 8, Not Upgrading: 42
Freed space: 7,496 kB
Continue? [Y/n] y
(Reading database ... 167229 files and directories currently installed.)
Removing cuyo-data (2.1.0-1) ...
Removing libfmt10:amd64 (10.1.1+ds1-4) ...
Removing libsdl2-mixer-2.0-0:amd64 (2.8.1+dfsg-2) ...
Removing libopusfile0:amd64 (0.12-4+b3) ...
Removing libphysfs1:amd64 (3.0.2-6+b2) ...
Removing libsdl2-image-2.0-0:amd64 (2.8.8+dfsg-1+deb13u1) ...
Removing libsdl2-ttf-2.0-0:amd64 (2.24.0+dfsg-2) ...
Removing libxmp4:amd64 (4.6.2-2) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for libc-bin (2.41-12+deb13u4) ...
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 42
ogsystem@cis106-desktop:~$

Question 2


Open
lab4-script1.sh
~/scripts
1 !#/bin/bash
1/1
Tilix: ogsystem@cis106-desktop: ~
1: ogsystem@cis106-desktop: ~
ogsystem@cis106-desktop:~$ bash -/scripts/lab4-script1.sh


Open
lab4-script2.sh
~/scripts
lab4-script1.sh
lab4-script2.sh
1 #!/bin/bash
2 echo "Hello World"
3 echo -e "Name:\tJoe"
4 echo -e "Age:\t35"
5 echo -e "Color:\tBlue\nGame:\tTetris"


1/1
Tilix: ogsystem@cis106-desktop: ~
1: ogsystem@cis106-desktop: ~
--posix
--pretty-print
--rcfile
--restricted
--verbose
--version
Shell options:
-ilrsD or -c command or -O shopt_option (invoc
ation only)
-abefhkmnptuvxBCEHPT or -o option
ogsystem@cis106-desktop:~$ bash ~/scripts/lab4-script2.sh
Hello World
ogsystem@cis106-desktop:~$ man echo
ogsystem@cis106-desktop:~$ bash ~/scripts/lab4-script2.sh
Hello World
Name: Joe
Age: 35
ogsystem@cis106-desktop:~$ bash ~/scripts/lab4-script2.sh
Hello World
Name: Joe

```
Age:    35
Color:  Blue
Game:   Tetris
```

Open lab4-script3.sh ~/scripts

```
1 #!/bin/bash
2 echo "Today's date is: "
3 date
4 echo -n "The current user is: "
5 whoami
6 # echo -n "The current user is: "; whoami
7 echo "Memory Usage: "
8 free -h
```

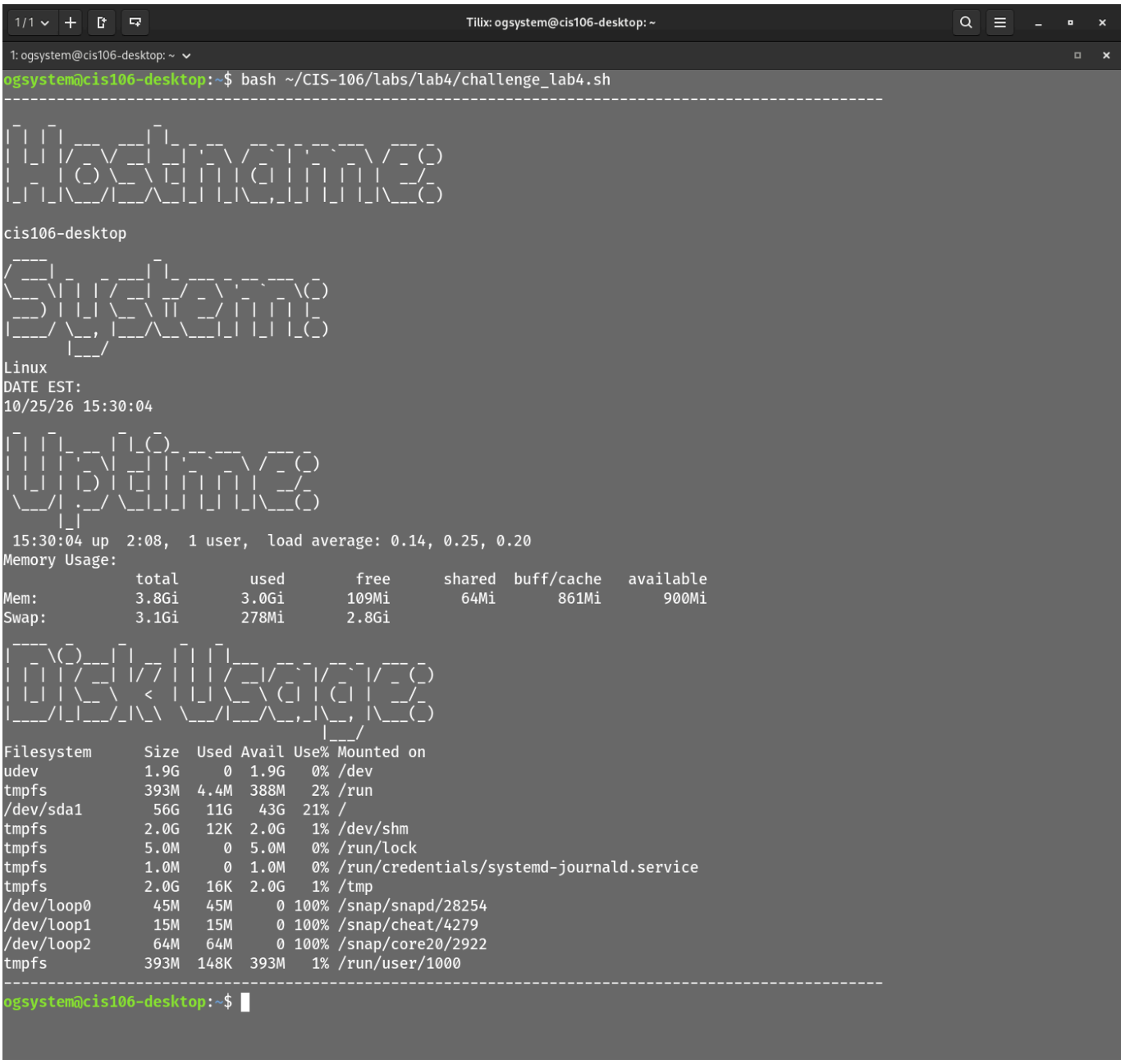
1/1 Tilix: ogsystem@cis106-desktop: ~

1: ogsystem@cis106-desktop: ~

```
Sun Oct 25 02:27:05 PM EDT 2026
The current user is: ogsystem
ogsystem@cis106-desktop:~$ bash ~/scripts/lab4-script3.sh
Today's date is:
Sun Oct 25 02:27:49 PM EDT 2026
The current user is: ogsystem
The current user is: ogsystem
ogsystem@cis106-desktop:~$ bash ~/scripts/lab4-script3.sh
Today's date is:
Sun Oct 25 02:28:45 PM EDT 2026
The current user is: ogsystem
Memory Usage:
-h
ogsystem@cis106-desktop:~$ bash ~/scripts/lab4-script3.sh
Today's date is:
Sun Oct 25 02:29:01 PM EDT 2026
The current user is: ogsystem
Memory Usage:
               total        used        free      shared  buff
```


/cache available
Mem: 3.8Gi 3.0Gi 114Mi 69Mi
933Mi 900Mi
Swap: 3.1Gi 180Mi 2.9Gi
ogsystem@cis106-desktop:~$

Question 3


1/1
Tilix: ogsystem@cis106-desktop: ~
1: ogsystem@cis106-desktop: ~
ogsystem@cis106-desktop:~$ bash ~/CIS-106/labs/lab4/challenge_lab4.sh
cis106-desktop
Linux
DATE EST:
10/25/26 15:30:04
15:30:04 up 2:08, 1 user, load average: 0.14, 0.25, 0.20
Memory Usage:
total used free shared buff/cache available
Mem: 3.8Gi 3.0Gi 109Mi 64Mi 861Mi 900Mi
Swap: 3.1Gi 278Mi 2.8Gi
Filesystem Size Used Avail Use% Mounted on
udev 1.9G 0 1.9G 0% /dev
tmpfs 393M 4.4M 388M 2% /run
/dev/sda1 56G 11G 43G 21% /
tmpfs 2.0G 12K 2.0G 1% /dev/shm
tmpfs 5.0M 0 5.0M 0% /run/lock
tmpfs 1.0M 0 1.0M 0% /run/credentials/systemd-journald.service
tmpfs 2.0G 16K 2.0G 1% /tmp
/dev/loop0 45M 45M 0 100% /snap/snapd/28254
/dev/loop1 15M 15M 0 100% /snap/cheat/4279
/dev/loop2 64M 64M 0 100% /snap/core20/2922
tmpfs 393M 148K 393M 1% /run/user/1000
ogsystem@cis106-desktop:~$

Open challange_lab4.sh ~/CIS-106/labs/lab4

lab4-script3.sh challange_lab4.sh

```
#!/bin/bash
length=100

for ((i=0; i<length; i++)); do
    printf "-"
done
printf "\n"


figlet "Hostname:"
hostname


figlet "System:"
uname

echo "DATE EST: "
date +"%D %T"


figlet "Uptime:"
uptime


echo "Memory Usage:"
free -h


figlet "Disk:"
df -h

for ((i=0;i<100;i++)); do
    printf "-"
done
printf "\n"
```

